

Paper Information Record — Stroube2025

Paper ID: 0005 | Oxford Missing Data Study

General Information

Field	Value
Paper ID	0005
Title	Mapping Entrepreneurial Inclusion Across US Neighborhoods: The Case of Low-Code E-commerce Entrepreneurs
Authors	Bryan Stroube, Gary Dushnitsky

Data Structure

Field	Value
Data type	Cross-section (ZCTA-level)
Observations (N)	32,647
Preferred model column	Table 3, Column 1 (Model 3_1)

Core Regression Model

Field	Value
Dependent variable	log_shopify_count_1
Focal IV	log_pop_black_aa
Controls	log_pop_total, log_total_bachelor_deg, log_pop_total_poverty, log_total_social_cap
Fixed effects	state_name_fe + MSA_fe (two-way, absorbed via pyfixest)
Clustered SEs	CRV1 on MSA_fe

Simulation Configuration

Field	Value
Key variables (4)	log_pop_black_aa, log_total_bachelor_deg, log_pop_total_poverty, log_total_social_cap
MAR control	log_pop_total
Mechanisms	MCAR, MAR, NMAR
Missingness proportions	1%, 5%, 10%, 20%, 30%, 40%, 50%
Methods	LD, Mean, Reg, Iter, RF, DL, MILGBM
Iterations per scenario	30
MAR/NMAR strength	1.5
M (multiple imputation)	5

Baseline Validation

Field	Value
Baseline coefficient	0.0307 (log_pop_black_aa)
Baseline SE	0.0054

Field	Value
Baseline p-value	<0.001 (p = 2.80e-08)
Published value	0.0307 (exact match)
R-squared	0.688

Run Log

Field	Value
Full run start	2026-03-29 21:17
DL method	TensorFlow 2.20.0, MLP 32->Dropout(0.1)->16->1, ReLU, Adam lr=0.005, EarlyStopping
MILGBM method	LightGBM 4.6.0, MICE M=5, Rubin's Rules, 30 estimators, max_depth=4